HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Installed and commissioned **semiconductor manufacturing equipment** in a **world-class cleanroom** environment, adhering strictly to OEM specifications and site protocols.
- Performed preventative maintenance and calibration on **sophisticated capital equipment**, contributing to a sustained **92% operational uptime** across the production line.
- Troubleshot and diagnosed complex **electrical, mechanical, and automation faults** on advanced manufacturing systems, performing root cause analysis for sustainable technical solutions.
- Supported equipment upgrades and continuous improvement initiatives, collaborating with process and equipment engineers to enhance **manufacturing yield**.
- Maintained detailed service documentation and technical reports, ensuring compliance with **ISO and semiconductor manufacturing standards**.
- Provided technical guidance to internal stakeholders and production teams, facilitating efficient equipment operation and troubleshooting.
- Participated in on-call support rotations, responding to critical manufacturing operations escalations and liaising with global engineering teams.

SKILLS

Engineering Discipline – Electrical Engineering, Electronic Engineering, Automation, Mechanical Engineering

Manufacturing Environment – Semiconductor Manufacturing, High-Tech Manufacturing, Cleanroom Protocols, GMP

Equipment Maintenance – Installation, Commissioning, Qualification, Preventative Maintenance, Calibration, Servicing

Troubleshooting & Analysis – Fault-finding, Root Cause Analysis (RCA), Diagnostic Tools, Technical Documentation

Software & Systems – Microsoft Office, Windows-based Software, EHS Procedures

Project & Compliance – Continuous Improvement Projects, Quality Compliance, ISO Standards, Technical Reporting

PROJECTS

Semiconductor Equipment Installation & Commissioning

- Managed the end-to-end installation and commissioning of a new line of **automated semiconductor processing units**, ensuring adherence to OEM specifications and site-specific cleanroom procedures.
- Developed and executed detailed test plans for equipment qualification, identifying and resolving minor electrical and mechanical integration issues prior to production handover, reducing initial setup time by **15%**.
- Collaborated with OEM technical support teams to resolve complex configuration challenges and ensure seamless integration into the existing manufacturing workflow.

Automated Equipment Maintenance & Optimization

- Designed and implemented a preventative maintenance schedule for a fleet of **critical wafer fabrication machines**, increasing mean time between failures (MTBF) by an estimated **20%**.
- Documented troubleshooting guides for common electrical and automation faults, enabling faster resolution times for the on-site support team.
- Performed detailed diagnostic analysis on underperforming equipment, identifying areas for process optimization to improve overall fab yield.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)